

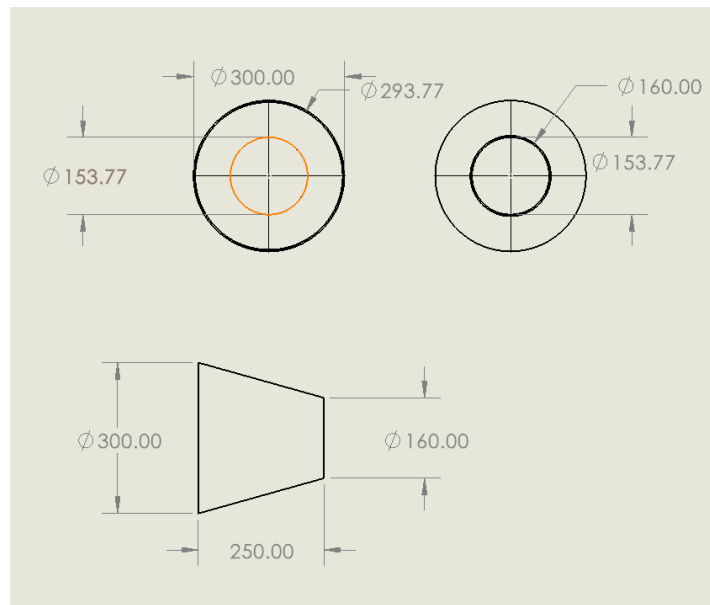
UNDERWATER DRONE NOZZLE (Reference)

1. Geometry Creation (SolidWorks)

Used SolidWorks to make the shape of the nozzle

Length of nozzle is 250mm.

Geometry exported in STEP/IGES format



2. Importing Geometry into ANSYS Workbench

Started ANSYS Workbench

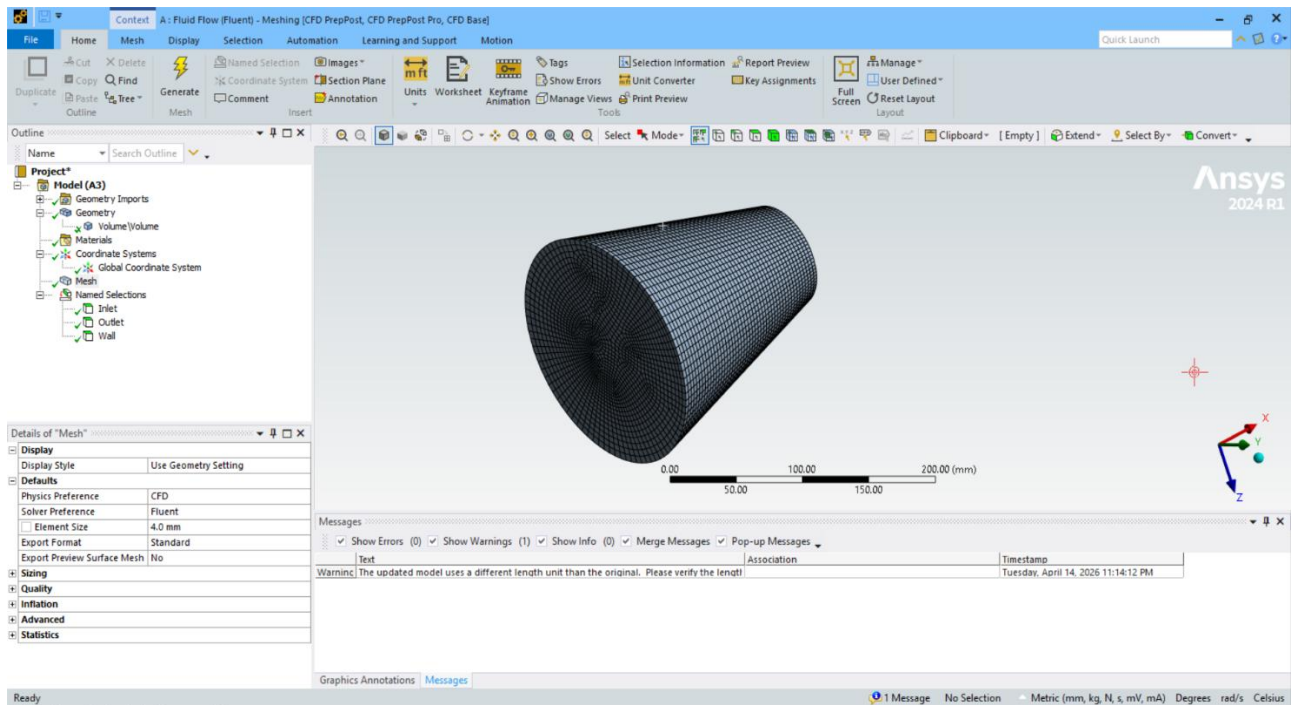
Dragged the fluid flow (Fluent) system

Imported geometry into the geometry module

3. Made a mesh

Boundaries with names:

Wall, inlet, and outlet



4. Settings for the Solver:

Solver: Based on Pressure

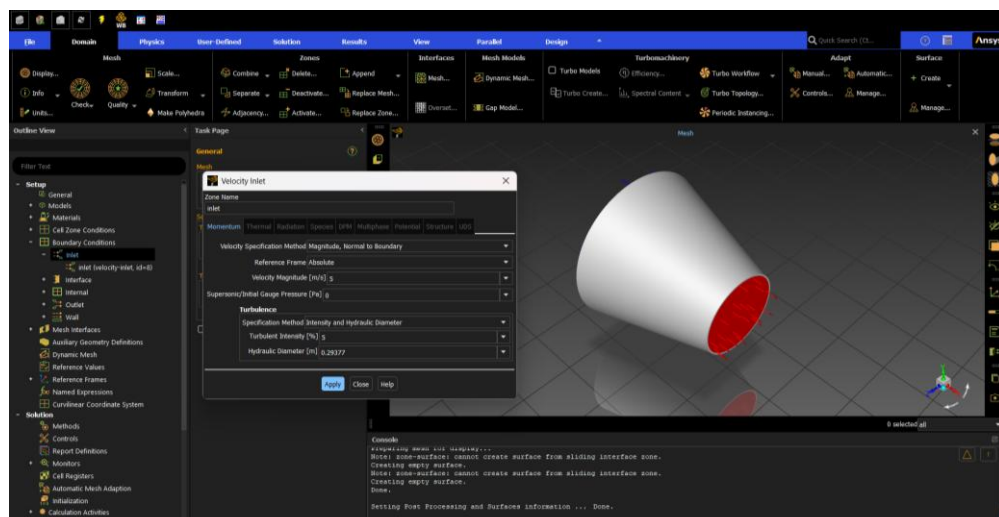
Flow that stays the same

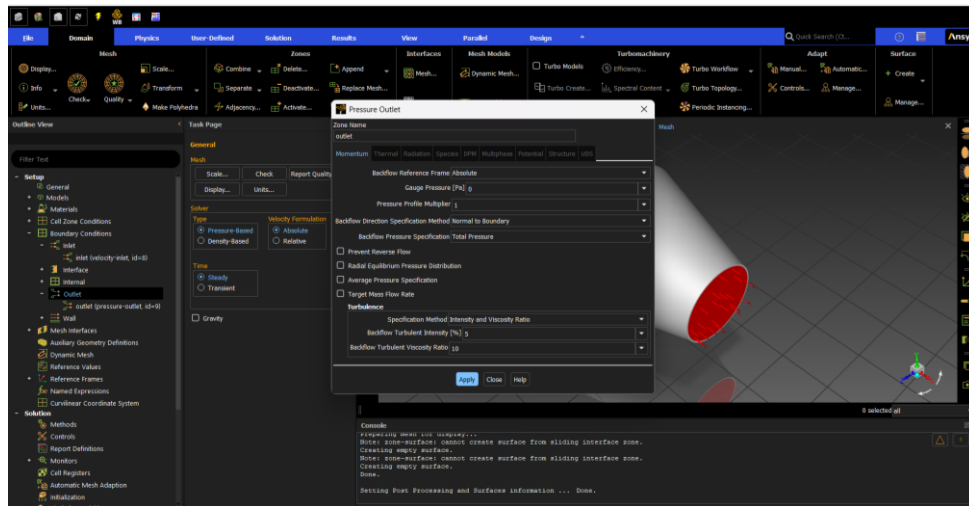
Water ($\rho = 998 \text{ kg/m}^3$) is the fluid.

Conditions at the Boundary:

Inlet $\rightarrow$ Velocity Inlet = 5 m/s

Outlet $\rightarrow$ Pressure Outlet (0 Pa gauge)





5. Solution (Run the Calculation)

Started the solution with Hybrid Initialization

Iterations: 50

6. Created a velocity contour

Seen:

Low speed at the inlet

High speed at the outlet

7. Results

- Inlet Velocity $\approx 5 \text{ m/s}$
- Outlet Velocity $\approx 19.15 \text{ m/s}$

Amplification Factor = $V_{\text{out}} / V_{\text{in}}$

$$= 19.15 / 5$$

$$= 3.83$$

